OSINT Field Guide

Training at Forrest.one · Open Source Intelligence collection, analysis and reporting



The Forrest.one OSINT fox mascot.

How to use this guide

This is a working manual, not background reading. Every procedure in it was run against a live domain while the guide was written, and every picture is a real capture of the tool that produced it.

Work through it in order the first time. After that, use the tool reference table at the back and jump straight to what you need.

Two rules before anything else:

1. Write scope before you run a single tool. Every investigation gets its own folder and its own SCOPE.md.
2. Capture raw output to disk. Never retype or tidy a result. If a tool fails, the failure is part of the record.

The legal and ethical boundary

Open source intelligence means information that is already public. This guide stays inside that line and so should you.

You may	You may not
Query public records: DNS, WHOIS, certificates	Access anything behind a login
Read public pages and documents	Attempt to bypass authentication

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You may	You may not
Search public breach corpora already published	Buy or use private personal data
Look at metadata in public files	Use findings to harass or expose an individual

Two further cautions that matter in practice:

- Some lookups are not silent. Registration checks such as holehe can trigger a password reset email. The subject may notice. Record that in the report rather than pretending the collection was invisible.
- Public still means personal. A home address found in a public record is still a home address. Handle it the way you would want yours handled.

1. Set up the collection lab

Everything runs inside one disposable container. The target never sees your own address and nothing you investigate touches your desktop.

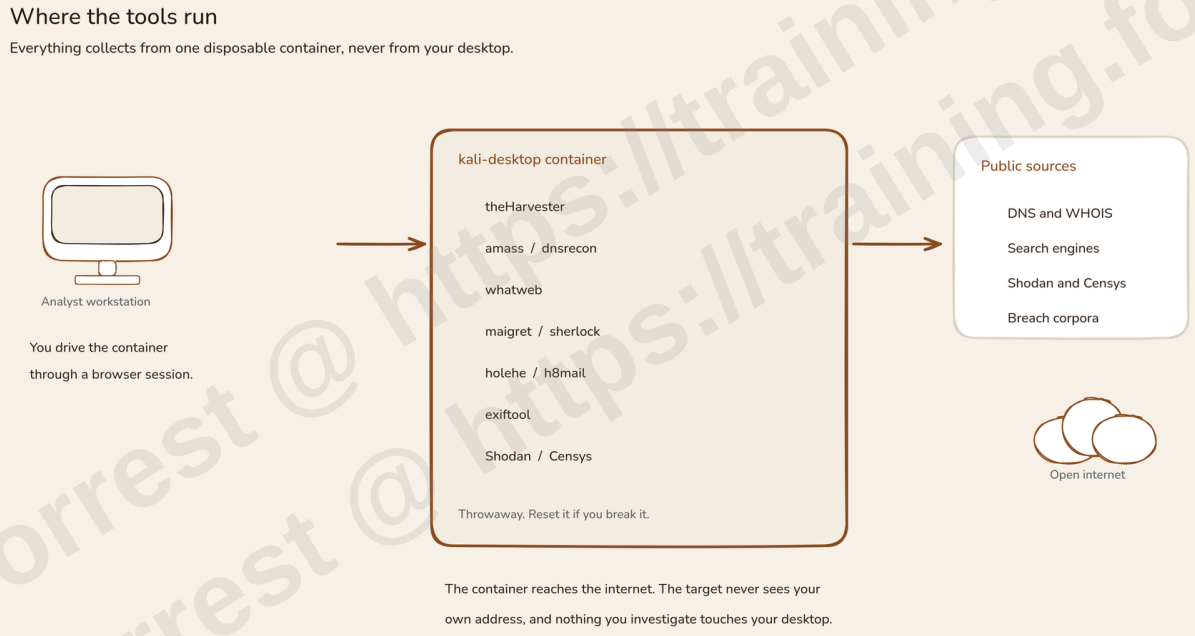


Figure 1: An analyst workstation driving a disposable collection container against public sources.

Why a container, not your desktop

- Attribution. Lookups arrive from the container's address, not yours.
- Safety. You will download unknown documents and visit unknown pages. A container you can delete contains that.
- Repeatability. The same tools at the same versions every time.

The environment facts that cost the most time

Four things break almost every first attempt. Check them before you blame a tool.

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Symptom	Cause	Fix
command not found for maigret, sherlock, holehe, h8mail, socialscan	pipx installs are not on the non-interactive PATH	export PATH=$PATH:/usr/local/bin inside the same bash -lc
List directory /var/lib/apt/lists/partial is missing	apt run as a non-root user	add -u 0 to docker exec
A tool reports a file missing that you just wrote	host and container see different filesystems	docker cp the file in
Output appears empty in your log	the tool wrote to stderr	capture with 2>&1

The wrapper every command uses

```
docker exec -u 0 kali-desktop bash -lc \
  'export PATH=$PATH:/usr/local/bin:/home/kasm-user/.local/bin; <command>'
```

-u 0 is required for anything touching apt. The PATH export is required for the pipx tools. Both together are the single most reliable command shape in this guide.

2. The workflow

Six phases. Infrastructure comes before people, because subdomains and mail hosts define the boundary for everything that follows.

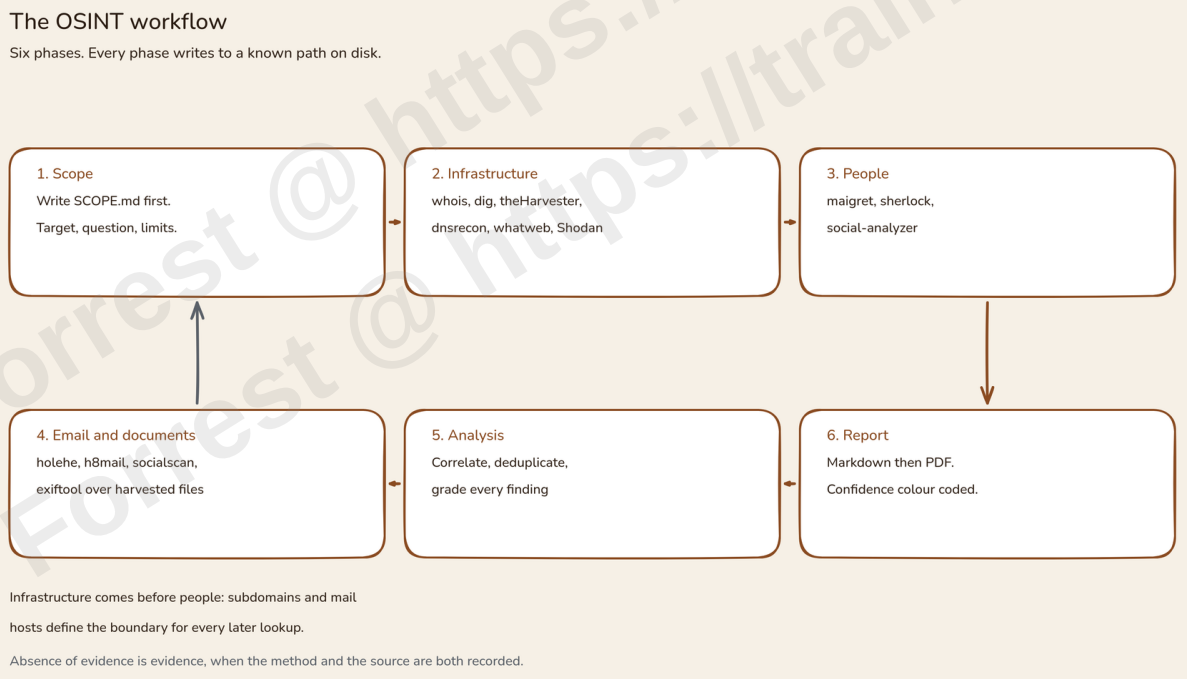


Figure 2: The six phase OSINT workflow, from scope to report.

Phase	Question it answers	Order
1. Scope	What am I looking at, and what am I allowed to do?	First, always
2. Infrastructure	What is registered, routable and internet-facing?	Then
3. People	Who works there, and where else do they exist?	After infrastructure

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Phase	Question it answers	Order
4. Email and documents	What is the address format, is it breached, what do files leak?	After people
5. Analysis	What corroborates, what is a coincidence, how confident am I?	After collection
6. Report	What does the reader need to know and act on?	Last

Why this order. Employee names feed the email format. Email addresses feed breach lookups. Breach data feeds document discovery. Run people collection first and you will guess the email format instead of deriving it.

3. Phase 1 — Scope

Before any tool runs, write the scope file. It exists so the run is documented, not so it can be approved.

```
# SCOPE — case-<slug>

- Target: forrest.one (domain)
- Question: what is publicly exposed under this domain?
- Collection: passive only. No port scanning, no login probing.
- Authorisation basis: standing authorisation, own asset.
- Out of scope: personal accounts of third parties.
- Started: 2026-10-01T21:40Z
```

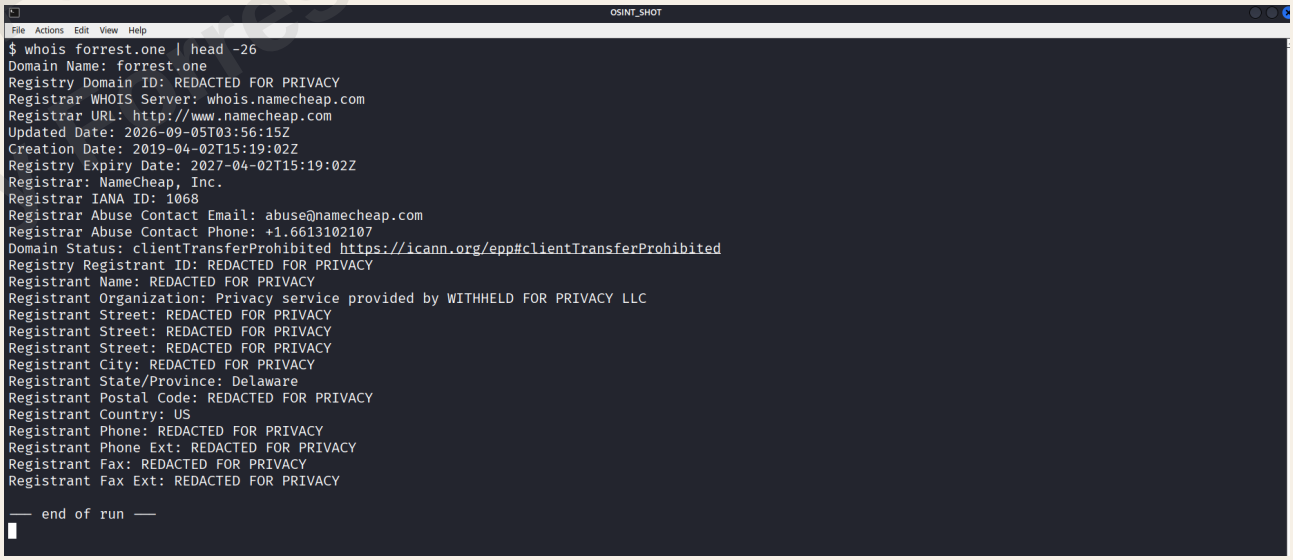
Four fields do the work: target, question, collection type, out of scope. An absent scope line is the most common reason a piece of collection has to be thrown away and redone.

4. Phase 2 — Infrastructure

4.1 WHOIS — who registered it, and when

WHOIS tells you the registrar, the creation and expiry dates, and the nameservers. On a privacy-protected domain it tells you that too, which is itself a finding.

```
docker exec -u 0 kali-desktop bash -lc 'whois forrest.one'
```



Screenshot: whois output for forrest.one, showing the registrar, creation date and privacy service.

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What the run above shows. Registration through NameCheap, created 2 April 2019, expiring 2 April 2027, with the registrant hidden behind a privacy service in Delaware, United States. Two years of paid-up registration on a privacy-protected record.

Read it like this:

Field	What it tells you
Creation Date	Domain age. A three-week-old domain in a phishing case is a signal.
Registry Expiry Date	Paid up or lapsing. Lapsing domains get dropped and re-registered.
Registrar	Where an abuse report would go.
Name Server	Where DNS actually lives. Cloudflare here means a proxy in front.
Registrant Organization	A privacy service is a finding, not a dead end.

Record the date of every WHOIS lookup. Public records change.

4.2 DNS — what exists and what routes it

Start broad, then narrow. ANY often returns nothing on modern nameservers; query the record types individually.

```
docker exec -u 0 kali-desktop bash -lc \
  'dig forrest.one ANY +noall +answer; dig +short MX forrest.one'
```

```
$ dig +short MX forrest.one
10 mx2.privateemail.com.
10 mx1.privateemail.com.

$ dig +short A forrest.one
104.21.30.130
172.67.172.239
```

Screenshot: dig returning the mail exchanger records for the domain.

What the run above shows. Two mail exchangers, mx1.privateemail.com and mx2.privateemail.com, each at priority 10. No A record through ANY, which is normal: many resolvers refuse ANY queries now.

Why the MX records matter. They name the mail provider, which tells you where an address format would be validated, and they are a third party that may hold useful hosted content.

4.3 dnsrecon — the full record set in one pass

Where dig answers one question at a time, dnsrecon enumerates the zone.

```
docker exec -u 0 kali-desktop bash -lc 'dnsrecon -d forrest.one -t std'
```

```
$ dnsrecon -d forrest.one -t std
2026-10-01T22:01:18.616071+0000 INFO     NS cleo.ns.cloudflare.com 173.245.59.89
2026-10-01T22:01:18.623541+0000 INFO     NS cleo.ns.cloudflare.com 172.64.33.89
2026-10-01T22:01:18.630201+0000 INFO     NS cleo.ns.cloudflare.com 2a06:98c1:50::ac40:2159
2026-10-01T22:01:18.630403+0000 INFO     NS cleo.ns.cloudflare.com 2803:f800:50::6ca2:c159
2026-10-01T22:01:18.630547+0000 INFO     NS cleo.ns.cloudflare.com 2606:4700:58::adf5:3b59
2026-10-01T22:01:18.630683+0000 INFO     NS tina.ns.cloudflare.com 108.162.192.230
2026-10-01T22:01:18.637581+0000 INFO     NS tina.ns.cloudflare.com 173.245.58.230
2026-10-01T22:01:18.644215+0000 INFO     NS tina.ns.cloudflare.com 172.64.32.230
2026-10-01T22:01:18.650846+0000 INFO     NS tina.ns.cloudflare.com 2606:4700:50::adf5:3ae6
2026-10-01T22:01:18.651033+0000 INFO     NS tina.ns.cloudflare.com 2a06:98c1:50::ac40:20e6
2026-10-01T22:01:18.651173+0000 INFO     NS tina.ns.cloudflare.com 2803:f800:50::6ca2:c0e6
2026-10-01T22:01:18.661132+0000 INFO     MX mx1.privateemail.com 162.255.118.7
2026-10-01T22:01:18.661193+0000 INFO     MX mx2.privateemail.com 162.255.118.8
2026-10-01T22:01:18.661945+0000 INFO     A forrest.one 172.67.172.239
2026-10-01T22:01:18.661996+0000 INFO     A forrest.one 104.21.30.130
2026-10-01T22:01:18.662025+0000 INFO     AAAA forrest.one 2606:4700:3031::6815:1e82
2026-10-01T22:01:18.662050+0000 INFO     AAAA forrest.one 2606:4700:3033::ac43:acef
2026-10-01T22:01:18.670644+0000 INFO     TXT forrest.one v=spf1 include:spf.privateemail.com ~all
2026-10-01T22:01:18.679166+0000 INFO Enumerating SRV Records
2026-10-01T22:01:18.710231+0000 ERROR No SRV Records Found for forrest.one
2026-10-01T22:01:18.710369+0000 INFO Completed enumeration for domain: forrest.one
```


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Screenshot: dnsrecon enumerating SOA, NS, MX, A, AAAA and TXT records.

What the run above shows. The complete public footprint in one view:

Type	Value	Reading
SOA / NS	cleo.ns.cloudflare.com, tina.ns.cloudflare.com	Authoritative DNS at Cloudflare
A	104.21.30.130, 172.67.172.239	Two proxy addresses, Cloudflare ranges
AAAA	2606:4700:3031::6815:1e82	IPv6 published
MX	mx1 / mx2.privateemail.com	Mail hosted off-site
TXT	v=spf1 include:spf.privateemail.com ~all	SPF present and correctly scoped
SRV	none found	No service discovery records published

Two findings worth naming. The Recursion enabled on NS Server errors are not your problem: they describe the nameserver's configuration, not the target's. And an SPF record ending ~all is a soft fail, which a spoofing assessment should note.

Absence is a finding. "No SRV records found" is a real result when you state the method. Put it in the report.

4.4 theHarvester — hosts, emails and names from public sources

theHarvester queries search engines and certificate sources for anything tied to the domain.

```
docker exec -u 0 kali-desktop bash -lc \
  'theHarvester -d forrest.one -b duckduckgo -l 300'
```

```
$ theHarvester -d forrest.one -b duckduckgo -l 300
* theHarvester 4.8.1                                  *
* Coded by Christian Martorella                       *
* Edge-Security Research                              *
* cmartorella@edge-security.com                       *
*                                                     *
*******************************************************************

[*] Target: forrest.one

[*] Searching Duckduckgo.

[*] No IPs found.

[*] No emails found.

[*] No people found.

[*] No hosts found.
```

Screenshot: theHarvester reporting no IPs, emails, people or hosts found.

What the run above shows a negative result, and a clean one. The tool exited normally, queried DuckDuckGo, and found nothing of any kind.

Do not treat this as a failure. It is a genuine finding with two parts:

1. Nothing about this domain is indexed by that engine under that source.
2. The method and source are recorded, so the negative is reproducible.

A run log that hides a null result is worth less than one that states it. Report negatives with the source named: "theHarvester with the DuckDuckGo source found no hosts for forrest.one on 1 October 2026."

If you need a positive result to demonstrate the workflow, run against a domain you control that has indexed content, or add sources with -b bing,duckduckgo,otx,crtsh and compare. Each source is a separate claim in the report.

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4.5 whatweb — what the site is actually running

```
docker exec -u 0 kali-desktop bash -lc 'whatweb -a 1 https://forrest.one'
```

```
$ whatweb -a 1 https://forrest.one
https://forrest.one [404 Not Found] Country[RESERVED][ZZ], HTTPServer[cloudflare], IP[172.67.172.239], Script, Title[404 Not Found], UncommonHeaders[x
bots-tag,nel,report-to,cf-cache-status,speculation-rules,cf-ray,alt-svc]
```

Screenshot: whatweb identifying Cloudflare, the HTTP status and response headers.

What the run above shows. A 404 at the root, served by Cloudflare, from 104.21.30.130, in the United States. Response headers include x-robots-tag, nel, report-to, cf-cache-status, speculation-rules, cf-ray and alt-svc.

Read the headers, not just the fingerprint. cf-ray confirms Cloudflare termination. x-robots-tag is a machine-readable instruction about indexing and is worth reading in full. The alt-svc header advertises alternative protocols.

Set the aggression level deliberately. -a 1 is a single polite request. Higher levels send more requests and are more **likely** to be logged or blocked.

4.6 Search operators — finding documents engines already indexed

Search engines have crawled far more of a target than you ever will by hand. Operators narrow that index to what you want.

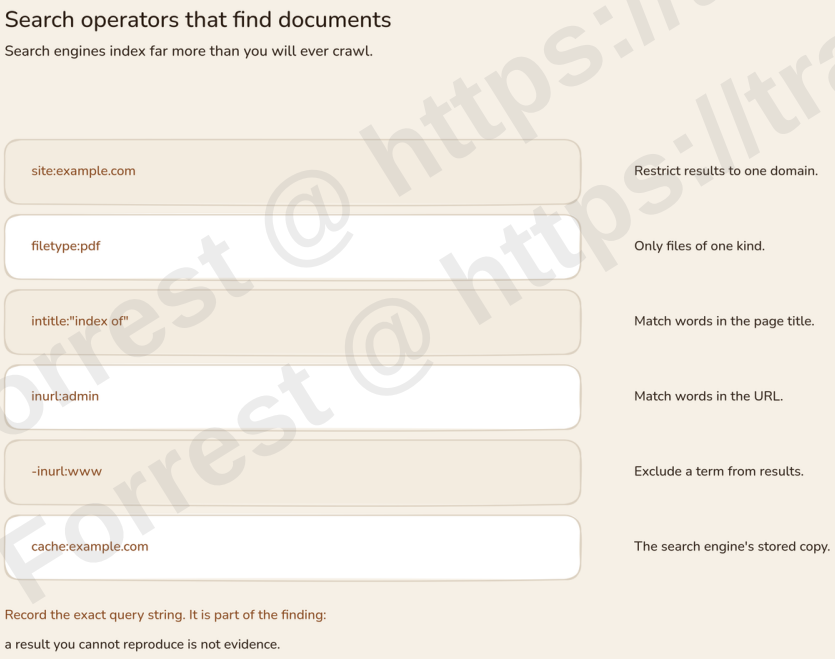


Figure 3: Search operators that find documents, with what each one does.

Operator	Finds
site:example.com	Everything indexed under one domain
filetype:pdf	Files of one kind
intitle:"index of"	Open directory listings
inurl:admin	Administrative paths referenced in URLs
-inurl:www	Excludes results carrying a term

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Operator	Finds
cache:example.com	The engine's stored copy of a page

Always record the exact query string. It is part of the finding. A result you cannot reproduce is not evidence.

go-dork automates the query syntax:

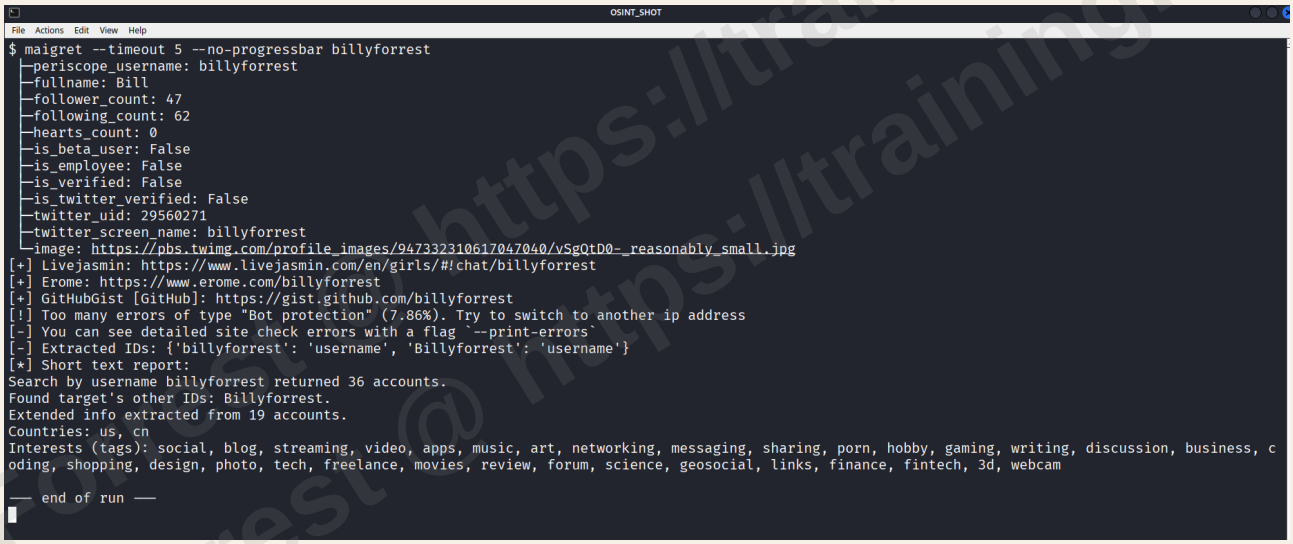
```
docker exec -u 0 kali-desktop bash -lc \
  'go-dork -q "site:forrest.one" -e google'
```

5. Phase 3 — People and usernames

5.1 maigret — one username, every platform

maigret takes a single username and checks it across hundreds of sites, returning not just hits but the profile data each platform exposes.

```
docker exec -u 0 kali-desktop bash -lc \
  'maigret --timeout 5 --no-progressbar billyforrest'
```



Screenshot: maigret reporting 36 accounts found for a username, with extracted identifiers.

What the run above shows. Thirty-six accounts for one username, extended information from nineteen of them, additional identifiers extracted, and a list of interests and tags.

This is the highest-yield step in the whole workflow, and the one to treat with most care. Three things to note:

- A username match is **Possible**, not **Confirmed**. People reuse usernames. Corroborate with a second attribute: a real name, a linked email, an avatar that appears elsewhere.
- Bot protection degrades results silently. The run logged Too many errors of type "Bot protection" (8.06%). That is a stated limitation of this run and belongs in the report.
- Extended profile data can include a linked social account and an avatar image URL. An avatar is a pivot: run it through a reverse image search.

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5.2 sherlock — the fast check

sherlock covers fewer sites than maigret but returns faster. Use it for a quick existence check, then maigret for the detail.

```
docker exec -u 0 kali-desktop bash -lc 'sherlock billyforrest'
```

5.3 social-analyzer — a graded confidence view

social-analyzer returns a confidence percentage per match rather than a binary hit, which is useful when a platform's answer is ambiguous.

```
docker exec -u 0 kali-desktop bash -lc \
  'social-analyzer --username billyforrest --metadata --output json'
```

6. Phase 4 — Email addresses and breach exposure

6.1 holehe — which services an address is registered with

```
docker exec -u 0 kali-desktop bash -lc 'holehe --only-used contact@forrest.one'
```



Screenshot: holehe checking 121 sites and reporting the registration summary for an address.

What the run above shows. 121 sites checked in 4.31 seconds, with the legend that governs how to read the result: [+] used, [-] not used, [x] rate limited. No services came back as used for this address.

Read the legend before you read the result. A [x] rate limit is not a negative. A run where most sites are rate limited has produced almost no information, and reporting it as "not registered anywhere" would be wrong.

Operational note. Several checks work by requesting a password reset. The subject can observe this. It is authorised on your own assets; on a third party it changes the risk profile entirely.

6.2 h8mail — breach corpus lookup

```
docker exec -u 0 kali-desktop bash -lc 'h8mail -t contact@forrest.one'
```

Breach data is only usable where it is already lawfully published. Treat a breach hit as **Possible** until you can corroborate it against something current: breach dumps contain stale passwords and recycled addresses.

6.3 socialscan — availability across platforms in one pass

```
docker exec -u 0 kali-desktop bash -lc \
  'socialscan billyforrest contact@forrest.one'
```


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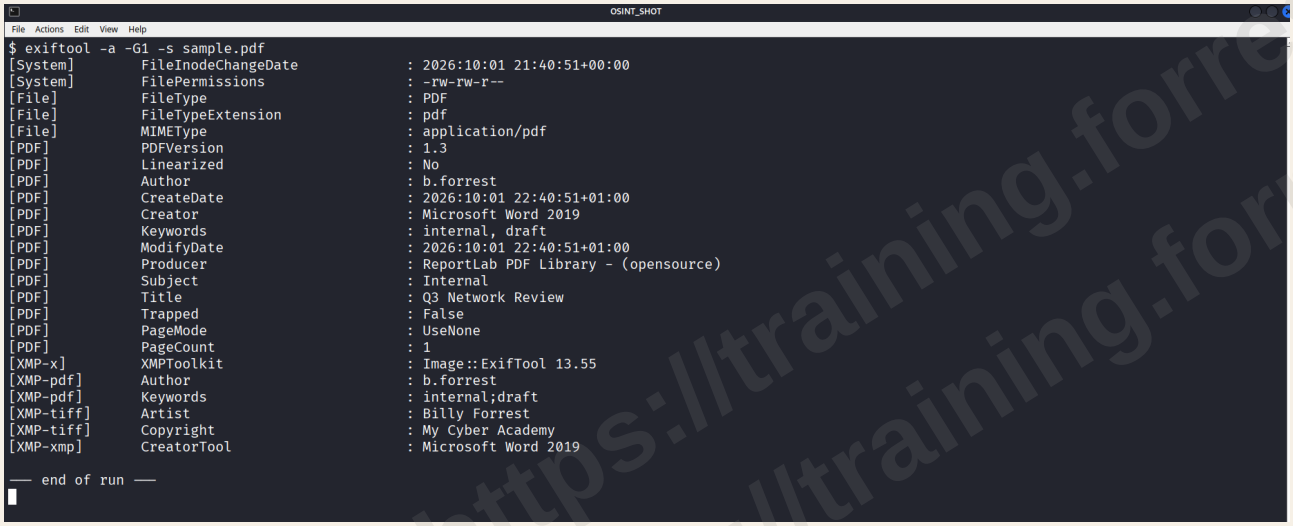
socialscan answers a different question from holehe: not whether an address is registered, but whether an address or username is available on each platform. Availability tells you which platforms the subject has not used, which narrows the field when a name is common.

7. Phase 5 — Documents and metadata

Documents leak more than people expect. A file's metadata can name its author, the software that made it, and for a photograph, the exact place it was taken.

7.1 Office and PDF documents

```
docker exec -u 0 kali-desktop bash -lc 'exiftool -a -G1 -s document.pdf'
```



Screenshot: exiftool showing PDF metadata including author, creator tool and keywords.

What the run above shows. The PDF carries an author name, a creator tool (Microsoft Word 2019), a title, a subject, keywords, and the producing application. The -G1 flag groups each tag by the metadata family it came from, and -s prints tag names rather than descriptions, which makes the output scriptable.

Three flags worth knowing:

Flag	Effect
-a	Show duplicate tags, including ones hidden in a lower-priority group
-G1	Group output by metadata family: PDF, XMP-pdf, XMP-tiff, IFD0
-s	Short tag names, so output can be piped and parsed

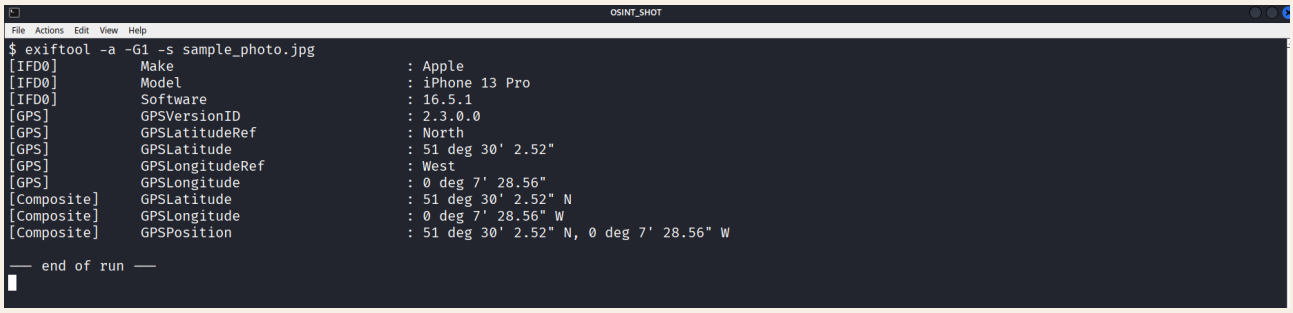
The XMP families are the leak. Note that the same author appears under [PDF], [XMP-pdf] and [XMP-tiff] in the run above. XMP survives edits that strip the simpler tags, so a document supposedly anonymised at the PDF level can still name its author in XMP.

7.2 Photographs — the GPS problem

```
docker exec -u 0 kali-desktop bash -lc \
  'exiftool -a -G1 -s photo.jpg | grep -iE "GPS|Model|Make|Software"'
```

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Screenshot: exiftool showing GPS position, camera make and model in a photograph.

What the run above shows. A camera make and model, a software version, a capture date, and a full GPS position in both decimal and degrees-minutes-seconds form.

This is the single most consequential metadata finding in OSINT. A photograph posted to a public page can carry the coordinates of the place it was taken. Two practical rules follow:

- When publishing, strip it. exiftool -all= file.jpg before a file leaves your organisation.
- When collecting, convert it. Run a GPS pair through a mapping service and record the human-readable location in the report, not just the raw numbers.

The same logic runs in reverse for theHarvester output. Point exiftool at every document the harvester gathered: one pass over a folder.

```
exiftool -r -a -G1 -s harvested/ | grep -iE "Author|Creator|GPS|Title"
```

> *metagoofil is not available. No PyPI distribution exists for it. Use exiftool over the documents theHarvester collected, which produces the same result.*

8. Phase 6 — Analysis and grading

Collection produces raw material. Analysis turns it into something a reader can act on, and the core of analysis is stating how confident you are.

Grading a finding

A claim without a level and a source is an opinion, not intelligence.

Confirmed	Two independent sources, or one authoritative record.	e.g. WHOIS record, certificate transparency
Likely	One strong source plus supporting context.	e.g. theHarvester host plus a live HTTP response
Possible	A single source that cannot be corroborated.	e.g. One social profile match
Unconfirmed	Reported by a source you cannot assess.	e.g. Third party claim, no direct evidence

State the level for every finding. Never upgrade a level without a second, independent source.

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Figure 4: The four confidence levels, what each one means and an example of evidence.

Level	Means	Evidence
Confirmed	Two independent sources, or one authoritative record	A WHOIS record; certificate transparency
Likely	One strong source plus supporting context	A harvested host plus a live HTTP response
Possible	One source that cannot be corroborated	A single social profile match
Unconfirmed	A source you cannot assess	A third-party claim with no direct evidence

Never upgrade a level without a second, independent source. Two lookups against the same underlying record are one source, not two. A WHOIS record and a dig answer from those nameservers are related, not independent.

The analysis steps

1. Deduplicate. One host reported by three tools in your lab is one finding.
2. Correlate. A username, an avatar and a linked email pointing at one person is a stronger claim than any one of them.
3. Separate fact from inference. "The MX record names privateemail.com" is fact. "The organisation therefore uses a paid mail provider" is inference. Label which is which.
4. Grade every finding. No level, no finding.
5. State the negatives. "Nothing found" with the method named.

9. Phase 7 — Reporting



Figure 5: From raw tool output to a graded report, through collection, evidence, analysis and reports folders.

Folder discipline

Every investigation gets its own case folder, and every phase writes to a known place.

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Folder	Holds
cases/case-<slug>/	The scope file, one folder per investigation
collection/	Raw, unedited, timestamped tool output
evidence/	Captured pages, documents, metadata dumps
analysis/	Working notes, correlation, deduplication
reports/	Finished reports and their markdown sources
exports/	Machine-readable output: CSV, JSON

Report structure

```
# OSINT Report — <target>

## Summary
Three sentences. What was asked, what was found, what it means.

## Scope and method
Target, question, collection window, tools used, what was out of scope.

## Findings
One table row per finding: statement, confidence level, source.

## Negative results
What was looked for and not found, with the method named for each.

## Limitations
Rate limiting, source coverage, anything that would change the result.

## Recommendations
What the reader should do next.
```

The findings table

Finding	Confidence	Source
Domain registered through NameCheap, created 2019	**Confirmed**	WHOIS, 1 Oct 2026
Authoritative DNS at Cloudflare	**Confirmed**	dnsrecon NS records
Mail hosted at privateemail.com	**Confirmed**	MX records
Registrant identity withheld behind a privacy service	**Confirmed**	WHOIS
No hosts indexed by DuckDuckGo	**Confirmed**	theHarvester, 1 Oct 2026
Username reused across several platforms	**Possible**	maigret, 36 matches

Verify before you deliver

```
pdfinfo report.pdf | grep -E 'Pages|Title|Author'
pdftotext report.pdf - | grep -c '<finding term>'
```

Count the findings in the PDF and compare with the findings table in the source. A report that renders is not the same as a report that is complete.

10. Tool reference

Every tool in this guide, with the invocation that works in the lab.

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Tool	Phase	Command
whois	Infrastructure	whois <domain>
dig	Infrastructure	dig +short MX <domain>
dnsrecon	Infrastructure	dnsrecon -d <domain> -t std
dnsenum	Infrastructure	dnsenum <domain>
fierce	Infrastructure	fierce --domain <domain>
amass	Infrastructure	amass enum -d <domain>
theHarvester	Infrastructure	theHarvester -d <domain> -b duckduckgo -l 300
whatweb	Infrastructure	whatweb -a 1 https://<domain>
shodan	Infrastructure	shodan host <ip>
censys	Infrastructure	censys search <query>
assetfinder	Infrastructure	assetfinder -subs-only <domain>
waybackurls	Infrastructure	waybackurls <<< <domain>
go-dork	Infrastructure	go-dork -q "<dork>" -e google
recon-ng	Infrastructure	recon-ng (interactive)
spiderfoot	Infrastructure	spiderfoot -s <domain>
photon	Infrastructure	photon -u <url>
maigret	People	maigret --timeout 5 --no-progressbar <username>
sherlock	People	sherlock <username>
social-analyzer	People	social-analyzer --username <name> --metadata
holehe	Email	holehe --only-used <email>
h8mail	Email	h8mail -t <email>
socialscan	Email	socialscan <query>
exiftool	Documents	exiftool -a -G1 -s <file>
nmap	Authorised only	nmap -sV <target>

Installing a missing tool

Tools are added to the disposable lab, not to your desktop. Three install paths cover everything above.

```
# Python CLI tools — PEP-668 blocked, so use pipx
pipx install maigret

# Go tools — export GOBIN so the binary lands on PATH
export GOPATH=/home/kasm-user/go GOBIN=/usr/local/bin
go install github.com/tomnomnom/assetfinder@latest

# A standalone script with no package
git clone --depth 1 https://github.com/s0md3v/Photon /opt/Photon
printf '#!/bin/bash\nexec python3 /opt/Photon/photon.py "$@"\n' > /usr/local/bin/photon
chmod +x /usr/local/bin/photon
```

Always smoke-test on a real input before declaring a tool installed. command -v <tool> proves the file exists, nothing more.

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```
assetfinder -subs-only example.com     # must return output, not just exist
```

Two install traps worth knowing:

- pip install photon installs a library, not the Photon tool. It has no command-line entry point. Clone the repository and write a launcher.
- go install fails for PhoneInfoga. Its build embeds a Vue frontend that needs node. Use the prebuilt release binary instead.

11. Troubleshooting

These are real failures hit while writing this guide, with the fix.

Symptom	Cause	Fix
theHarvester finds nothing on every source	engines returning no results for a **low**-profile domain	run additional sources with -b bing,duckduckgo,otrs,crtsh and report each separately
waybackurls returns nothing	a young domain with no archived history	state it as a finding; try curl against the Wayback CDX API directly
maigret logs Too many errors of type "Bot protection"	target platforms rate limiting the container	lower concurrency, or accept the reduced coverage and state the percentage
holehe shows [x] for most sites	rate limits, not absence	do not report rate-limited sites as negative
whois output is all REDACTED FOR PRIVACY	registrar privacy service in use	that is the finding; pursue other pivots
A metadata field appears three times	the same value stored in PDF, XMP-pdf and XMP-tiff	expected; report the XMP copy as the durable one
Output is empty in your log but the tool ran	output went to stderr	capture with 2>&1

12. Appendix — reproducing the screenshots

The images in this guide are captures of real terminal sessions, not reconstructions. The capture tool and the commands are included so any figure can be regenerated.

```
# 1. Run the real command in the lab, keeping the transcript
docker exec -u 0 kali-desktop bash -lc \
  'export PATH=$PATH:/usr/local/bin:/home/kasm-user/.local/bin; <command>' \
  > evidence/$(date -u +%Y%m%dT%H%M%SZ)_<slug>.txt 2>&1

# 2. Screenshot the terminal that produced it
tools/term_shot.py images/<slug>.png <seconds> -- '<command>'
```

Pitfalls the capture tool exists to handle — each one cost real time:

- qterminal has no title flag. Set the window title with a printf escape and locate the window with xdotool search --name.
- The VNC screen starts at 1024x768. Run xrandr -s 1920x1080 first or the capture is a small letterboxed desktop.
- Write the command to a script file inside the container. Inlining it crosses four quoting layers and the escape sequences are silently eaten.

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- Use pkill -x qterminal, not pkill -f qterminal. The -f form matches the docker exec command line and kills the caller.
- Raise the terminal font before capture. At the default size the output is unreadable once scaled into a document.

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